Appendix A5

A.5 The Multivariate Normal Distribution

A.5.1 The density function

The scalar random variable X has a normal (or Gaussian) distribution if its density function is given by

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \tag{A.1}$$

where μ and σ are scalars. It satisfies $\mathrm{E}(X) = \mu$ and $\mathrm{var}(X) = \sigma^2$, and we write

$$X \sim \mathcal{N}(\mu,\ \sigma^2). \tag{A.2}$$

Let Z be the standardized rv.:

$$Z = \frac{X - \mu}{\sigma},$$

then

$$f_Z(z) = f_X(\sigma Z + \mu) \left| \frac{\mathrm{d}z}{\mathrm{d}x} \right|^{-1} \tag{A.3}$$

$$= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(\sigma Z)^2}{2\sigma^2}} \sigma \tag{A.4}$$

$$= \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}, \tag{A.5}$$

and $\mathrm{E}(Z) = 0$, $\mathrm{var}(Z) = 1$ and $Z \sim \mathcal{N}(0,\ 1)$. Furthermore, X can be written as

$$X = \sigma Z + \mu. \tag{A.6}$$

The generalization to a multivariate context can be formulated based on a k vector $\boldsymbol{Z}$ with elements that are independent and identically distributed with a standard normal distribution $Z_1 \sim \mathcal{N}(0,\ 1)$.

As the elements of $\boldsymbol{Z}$ are independent we have

$$f_{\boldsymbol{Z}}(\boldsymbol{z}) = \prod_{i=1}^{k} f_{Z_i}(z_i) \tag{A.7}$$

$$= \prod_{i=1}^{k} \frac{1}{\sqrt{2\pi}} e^{-\frac{z_i^2}{2}} \tag{A.8}$$

$$= \frac{1}{(2\pi)^{k/2}} e^{-(1/2)\sum_{i=1}^{k} z_i^2} \tag{A.9}$$

$$= \frac{1}{(2\pi)^{k/2}} e^{-(1/2)\boldsymbol{z}'\boldsymbol{z}}, \tag{A.10}$$

where $\boldsymbol{z}$ is a $k \times 1$ vector, which is the point where the density is evaluated. We write

$$\boldsymbol{Z} \sim \mathcal{N}(\boldsymbol{0}_k,\ \boldsymbol{I}_k).$$

It has moments that agree with

$$\mathrm{E}(\boldsymbol{Z}) = \boldsymbol{0}: \quad k \times 1, \tag{A.11}$$

$$\mathrm{Var}(\boldsymbol{Z}) = \boldsymbol{I}_k: \quad k \times k. \tag{A.12}$$

Remark 1
Polar coordinates
Let $k = 2$, so $\boldsymbol{z} = (z_1, z_2)'$. Instead of the coordinates z_1 and z_2, we could also use so-called polar coordinates where r is the lenght of $\boldsymbol{z}$, i.e. $r = \sqrt{z_1^2 + z_2^2}$ and let ϕ be the angle between the vector $\boldsymbol{z}$ and the z_1 axis. So,

$$z_1 = r\cos\phi, \tag{A.13}$$

$$z_2 = r\sin\phi. \tag{A.14}$$

We find

$$f(r,\phi) = f_{\boldsymbol{Z}}(\boldsymbol{z})\left|\det\{\partial\boldsymbol{z}/\partial(r,\phi)\}\right|, \tag{A.15}$$

where the Jacian matrix is given by

$$\partial\boldsymbol{z}/\partial(r,\phi) = \begin{pmatrix} \cos\phi & -r\sin\phi \\ \sin\phi & r\cos\phi \end{pmatrix} \tag{A.16}$$

so, the Jacobian is given by

$$\left|\det\{\partial\boldsymbol{z}/\partial(r,\phi)\}\right| = \left|r\cos^2\phi + r\sin^2\phi\right| = r \tag{A.17}$$

The normal bivariate density $f_{\boldsymbol{z}}(z)$ could be formulated in terms of r and ϕ. Instead of $\boldsymbol{z}'\boldsymbol{z}$, we would write r^2, and instead of $\mathrm{d}z_1\,\mathrm{d}z_2$, we would write $r\,\mathrm{d}r\,\mathrm{d}\phi$.

Suppose we wouldn't know the scale factor $(2\pi)^{-1/2}$ of the univariate normal density and write for the bivariate density, formed by two independent standard normally distributed rvs, $f_{\boldsymbol{Z}}(\boldsymbol{z}) = a^2 e^{-(z_1^2+z_2^2)/2}$, to find the factor a we would integrate the function over the two-dimensional plane and choose the constant a such that the result would be one. Let's do it,

$$\int_{\mathbb{R}^2} f_{\boldsymbol{Z}}(\boldsymbol{z})\,\mathrm{d}\boldsymbol{z} = a^2 \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(z_1^2+z_2^2)/2}\,\mathrm{d}z_1\,\mathrm{d}z_2, \tag{A.18}$$

which is difficult to evaluate, but it equals in polar coordinates

$$= a^2 \int_0^{2\pi}\int_0^{\infty} e^{-r^2/2} r\,\mathrm{d}r\,\mathrm{d}\phi \tag{A.19}$$

$$= a^2 \int_0^{2\pi}\int_0^{\infty} e^{-u}\,\mathrm{d}u\,\mathrm{d}\phi \tag{A.20}$$

$$= a^2 \int_0^{2\pi} -e^{-u}\Big]_0^{\infty}\,\mathrm{d}\phi \tag{A.21}$$

$$= a^2 \int_0^{2\pi} \mathrm{d}\phi = a^2 2\pi. \tag{A.22}$$

So, $a = (2\pi)^{-1/2}$.

Remark 2
Spherical distributions
The multivariate standard normal distribution, i.e. when $\boldsymbol{Z} \sim \mathcal{N}(\mathbf{0}_k, \boldsymbol{I}_k)$*, is said to be a* spherical distribution*. The set of points* $\boldsymbol{z}$ *where the density is constant forms a multi-dimensional sphere. That is to say, the set* $\{\boldsymbol{z} \mid f_{\boldsymbol{Z}}(\boldsymbol{z}) = c\}$ *forms a shere. It is given by* $\{\boldsymbol{z} \mid \boldsymbol{z}'\boldsymbol{z} = d\}$ *for some* $d = -2\ln(c) - k\ln(2\pi)$*. It is a so-called* equidensity set *or* equidensity contour*. For spherical distributions the vectors in equidensity sets have equal lenght. In two dimensions the equidensity contours are circles.*

Now consider the random k vector

$$\boldsymbol{Y} = \boldsymbol{A}\boldsymbol{Z} + \boldsymbol{\mu}, \tag{A.23}$$

where $\boldsymbol{A}$ is a nonsingular nonrandom $k \times k$ matrix, and $\boldsymbol{\mu}$ is a nonrandom k vector. We know

$$\mathrm{E}(\boldsymbol{Y}) = \boldsymbol{\mu}, \tag{A.24}$$
$$\mathrm{Var}(\boldsymbol{Y}) = \boldsymbol{A}\,\mathrm{Var}(\boldsymbol{Z})\boldsymbol{A}' = \boldsymbol{A}\boldsymbol{A}'. \tag{A.25}$$

The density of $\boldsymbol{Y} = \boldsymbol{A}\boldsymbol{Z} + \boldsymbol{\mu}$ evaluated in a point $y = \boldsymbol{A}\boldsymbol{z} + \boldsymbol{\mu}$, so $\boldsymbol{z} = \boldsymbol{A}^{-1}(\boldsymbol{y} - \boldsymbol{\mu})$, and it is given by

$$f_{\boldsymbol{Y}}(y) = f_{\boldsymbol{Z}}\{\boldsymbol{A}^{-1}(\boldsymbol{y} - \boldsymbol{\mu})\} \left|\det\left(\frac{\partial \boldsymbol{z}}{\partial \boldsymbol{y}'}\right)\right| \tag{A.26}$$

where the last factor on the right hand side is the absolute value of the determinant of the Jacobian matrix $\partial \boldsymbol{z}/\partial \boldsymbol{y}'$.

We find

$$\begin{aligned}
\left|\det\left(\frac{\partial \boldsymbol{z}}{\partial \boldsymbol{y}'}\right)\right| &= \sqrt{\det\left(\boldsymbol{A}^{-1}\right)\det\left(\boldsymbol{A}^{-1}\right)} && \text{(A.27)}\\
&= \sqrt{\{\det\left(\boldsymbol{A}\right)\}^{-1}\{\det\left(\boldsymbol{A}\right)\}^{-1}} && \text{(A.28)}\\
&= \sqrt{\{\det\left(\boldsymbol{A}\right)\}^{-1}\{\det\left(\boldsymbol{A}'\right)\}^{-1}} && \text{(A.29)}\\
&= \sqrt{\{\det\left(\boldsymbol{A}\boldsymbol{A}'\right)\}^{-1}} && \text{(A.30)}\\
&= \{\det\left(\boldsymbol{A}\boldsymbol{A}'\right)\}^{-1/2} = \{\det(\boldsymbol{V})\}^{-1/2}, && \text{(A.31)}
\end{aligned}$$

where $\boldsymbol{V} = \mathrm{Var}(\boldsymbol{Y}) = \boldsymbol{A}\boldsymbol{A}'$.

Furthermore,

$$\boldsymbol{z}'\boldsymbol{z} = \left\{\boldsymbol{A}^{-1}(\boldsymbol{y} - \boldsymbol{\mu})\right\}'\boldsymbol{A}^{-1}(\boldsymbol{y} - \boldsymbol{\mu}) = (\boldsymbol{y} - \boldsymbol{\mu})'\boldsymbol{V}^{-1}(\boldsymbol{y} - \boldsymbol{\mu}). \tag{A.32}$$

We thus find

$$\begin{aligned}
f_{\boldsymbol{Y}}(\boldsymbol{y}) &= \frac{1}{(2\pi)^{k/2}\{\det\left(\boldsymbol{V}\right)\}^{1/2}} e^{-(1/2)(\boldsymbol{y}-\boldsymbol{\mu})'\boldsymbol{V}^{-1}(\boldsymbol{y}-\boldsymbol{\mu})} && \text{(A.33)}\\
&= \left\{\det(2\pi\boldsymbol{V})e^{(\boldsymbol{y}-\boldsymbol{\mu})'\boldsymbol{V}^{-1}(\boldsymbol{y}-\boldsymbol{\mu})}\right\}^{-1/2}. && \text{(A.34)}
\end{aligned}$$

We write

$$\boldsymbol{Y} \sim \mathcal{N}(\boldsymbol{\mu},\ \boldsymbol{V}).$$

This definition is in agreement with the univariate distribution such that, e.g.

$$\boldsymbol{a}'\boldsymbol{Y} \sim \mathcal{N}(\boldsymbol{a}'\boldsymbol{y},\ \boldsymbol{a}'\boldsymbol{V}\boldsymbol{a}), \tag{A.35}$$

since $\boldsymbol{a}'\boldsymbol{Y}$ can be written as $\boldsymbol{a}'(\boldsymbol{A}\boldsymbol{Z}+\boldsymbol{\mu}) = \boldsymbol{a}'\boldsymbol{A}\boldsymbol{Z}+\boldsymbol{a}'\boldsymbol{\mu}$, which has mean $\boldsymbol{a}'\boldsymbol{\mu}$ and variance $\boldsymbol{a}'\boldsymbol{V}\boldsymbol{a}$.

So, linear transformations of normally distributed vectors are again normally distributed. We only need the expectation and the covariance matrix to determine the density:

$$\boldsymbol{X} \sim \mathcal{N}\left(\mathrm{E}(\boldsymbol{X}),\ \mathrm{Var}(\boldsymbol{X})\right). \tag{A.36}$$

In fact, we could also say that $\boldsymbol{X}$ has a multivariate normal distribution if any linear combination $\boldsymbol{Y} = \boldsymbol{a}'\boldsymbol{X}$ has a univariate normal distriution. In that case $\boldsymbol{Y}$ is said to be normally distributed even if its covariance matrix is singular.

Remark 3
Elliptical distributions and principal axes
The general multivariate normal distribution, where $\boldsymbol{Y} \sim \mathcal{N}(\boldsymbol{\mu},\ \boldsymbol{V})$ and $\boldsymbol{V}$ is nonsigular, i.e. $\boldsymbol{V} > 0$, is said to be an elliptical distribution. *We say that if the equidensity sets are ellipsoids $\mathcal{E}$ given by*

$$\mathcal{E} = \left\{\boldsymbol{y} \mid (\boldsymbol{y}-\boldsymbol{\mu})'\boldsymbol{V}^{-1}(\boldsymbol{y}-\boldsymbol{\mu}) = c\right\}.$$

The ellipsoid $\mathcal{E}$ has center $\boldsymbol{\mu}$. In two dimensions, $k = 2$, it is an ellips. It has two axes. These axes are called principal axes and their directions are directly related the eigenvectors of $\boldsymbol{V}$.

To find such a direction, let the eigenvalue-vector decomposition of $\boldsymbol{V}$ be given by $\boldsymbol{V} = \boldsymbol{P}\boldsymbol{\Lambda}\boldsymbol{P}'$, where $\boldsymbol{P}\boldsymbol{P}' = \boldsymbol{P}'\boldsymbol{P} = \boldsymbol{I}_k$ and $\boldsymbol{\Lambda}$ is diagonal with elements $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_k > 0$. Write $\boldsymbol{y} = \boldsymbol{\mu} + \boldsymbol{P}\boldsymbol{x}$, so the length of $\boldsymbol{y}-\boldsymbol{\mu}$ is the length of $\boldsymbol{x}$. In terms of $\boldsymbol{x}$ we have

$$\boldsymbol{x}'\boldsymbol{\Lambda}^{-1}\boldsymbol{x} = \frac{x_1^2}{\lambda_1} + \frac{x_2^2}{\lambda_2} = c \tag{A.37}$$

The longest $\boldsymbol{x}$ is found for $x_1 = \sqrt{c\lambda_1}$ and $x_2 = 0$, which amounts to $\boldsymbol{y} = \boldsymbol{\mu} + \boldsymbol{p}_1\sqrt{c\lambda_1}$.

Remark 4
Independence and uncorrelatedness
Notice that if the $k \times 1$ vector $\boldsymbol{Y}$ is normally distributed with diagonal covariance matrix, so that the elements Y_i, $i = 1, \ldots, k$ are uncorrelated, then

$$f_{\boldsymbol{Y}}(\boldsymbol{y}) = \prod_{i=1}^{k} f_{Y_i}(y_i). \tag{A.38}$$

So, as the multivariate density factorizes in the univariate densities, the random variables Y_i, $i = 1, \ldots, k$ are independent.

So, under multivariate normality, uncorrelatedness is equivalent to independence.

In general, independence implies uncorrelatedness, but the reverse is not true.

Also notice that if Y_1 and Y_2 have marginal normal distributions, then the vector $\boldsymbol{Y} = (Y_1, Y_2)'$ need not have a bivariate normal distribution. This holds true even if Y_1 and Y_2 are uncorrelated. Check Exercise A7 on p. 311 of the book.

A.5.2 The characteristic function

As we know, using $\boldsymbol{A} = \boldsymbol{V}^{-1}$,

$$\int_{\mathbb{R}^k} e^{-\frac{1}{2}(\boldsymbol{y}-\boldsymbol{\mu})'\boldsymbol{A}(\boldsymbol{y}-\boldsymbol{\mu})}\,\mathrm{d}\boldsymbol{y} = (2\pi)^{k/2}\{\det(\boldsymbol{A})\}^{-1/2}, \tag{A.39}$$

we can easily compute integrals of the form

$$\int_{\mathbb{R}^k} e^{-\boldsymbol{y}'\boldsymbol{A}\boldsymbol{y}+\boldsymbol{b}'\boldsymbol{y}}\,\mathrm{d}\boldsymbol{y}, \tag{A.40}$$

where $\boldsymbol{A} > 0$.

That is to say,

$$(\boldsymbol{y} - \boldsymbol{A}^{-1}\boldsymbol{b}/2)'\boldsymbol{A}(\boldsymbol{y} - \boldsymbol{A}^{-1}\boldsymbol{b}/2) = \boldsymbol{y}'\boldsymbol{A}\boldsymbol{y} - \boldsymbol{b}'\boldsymbol{y} + \boldsymbol{b}'\boldsymbol{A}^{-1}\boldsymbol{b}/4, \tag{A.41}$$

So, if $\boldsymbol{A} > 0$,

$$\begin{aligned}
\int_{\mathbb{R}^k} e^{-\boldsymbol{y}'\boldsymbol{A}\boldsymbol{y}+\boldsymbol{b}'\boldsymbol{y}}\,\mathrm{d}\boldsymbol{y} &= e^{\frac{1}{4}\boldsymbol{b}'\boldsymbol{A}^{-1}\boldsymbol{b}} \int_{R^k} e^{-(\boldsymbol{y}-\boldsymbol{A}^{-1}\boldsymbol{b}/2)'\boldsymbol{A}(\boldsymbol{y}-\boldsymbol{A}^{-1}\boldsymbol{b}/2)}\,\mathrm{d}\boldsymbol{y} && \text{(A.42)}\\
&= e^{\frac{1}{4}\boldsymbol{b}'\boldsymbol{A}^{-1}\boldsymbol{b}} \int_{\mathbb{R}^k} e^{-\frac{1}{2}(\boldsymbol{y}-\boldsymbol{A}^{-1}\boldsymbol{b}/2)'(2\boldsymbol{A})(\boldsymbol{y}-\boldsymbol{A}^{-1}\boldsymbol{b}/2)}\,\mathrm{d}\boldsymbol{y} && \text{(A.43)}\\
&= e^{\frac{1}{4}\boldsymbol{b}'\boldsymbol{A}^{-1}\boldsymbol{b}}(2\pi)^{k/2}\{\det(2\boldsymbol{A})\}^{-1/2} && \text{(A.44)}\\
&= e^{\frac{1}{4}\boldsymbol{b}'\boldsymbol{A}^{-1}\boldsymbol{b}}\pi^{k/2}\{\det(\boldsymbol{A})\}^{-1/2}, && \text{(A.45)}
\end{aligned}$$

which is a result similar to Lemma A.5.3 in the book.

Among such integrals we find the moment generating function of the normal distribution. Let $\boldsymbol{Y} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{V})$, then the moment generating function is given by

$$\begin{aligned}
\mathrm{E}\left(e^{\boldsymbol{t}'\boldsymbol{Y}}\right) &= (2\pi)^{-k/2}\{\det(\boldsymbol{V})\}^{-1/2} \int_{\mathbb{R}^k} e^{\boldsymbol{t}'\boldsymbol{y}} e^{-\frac{1}{2}(\boldsymbol{y}-\boldsymbol{\mu})'\boldsymbol{V}^{-1}(\boldsymbol{y}-\boldsymbol{\mu})}\,\mathrm{d}\boldsymbol{y} && \text{(A.46)}\\
&= (2\pi)^{-k/2}\{\det(\boldsymbol{V})\}^{-1/2} \int_{\mathbb{R}^k} e^{-\frac{1}{2}(\boldsymbol{y}-\boldsymbol{\mu})'\boldsymbol{V}^{-1}(\boldsymbol{y}-\boldsymbol{\mu})+\boldsymbol{t}'\boldsymbol{y}}\,\mathrm{d}\boldsymbol{y} && \text{(A.47)}\\
&= (2\pi)^{-k/2}\{\det(\boldsymbol{V})\}^{-1/2} e^{-\frac{1}{2}\boldsymbol{\mu}'\boldsymbol{V}^{-1}\boldsymbol{\mu}} \int_{\mathbb{R}^k} e^{-\boldsymbol{y}'(2\boldsymbol{V})^{-1}\boldsymbol{y}+(\boldsymbol{t}+\boldsymbol{V}^{-1}\boldsymbol{\mu})'\boldsymbol{y}}\,\mathrm{d}\boldsymbol{y}, && \text{(A.48)}
\end{aligned}$$

and using $\boldsymbol{b} = \boldsymbol{t} + \boldsymbol{V}^{-1}\boldsymbol{\mu}$ and $\boldsymbol{A} = (2\boldsymbol{V})^{-1}$ we find

$$\mathrm{E}\left(e^{\boldsymbol{t}'\boldsymbol{Y}}\right) = (2\pi)^{-k/2}\{\det(\boldsymbol{V})\}^{-1/2}e^{-\frac{1}{2}\boldsymbol{\mu}'\boldsymbol{V}^{-1}\boldsymbol{\mu}} \quad \times \tag{A.49}$$

$$e^{\frac{1}{4}(\boldsymbol{t}+\boldsymbol{V}^{-1}\boldsymbol{\mu})'2\boldsymbol{V}(\boldsymbol{t}+\boldsymbol{V}^{-1}\boldsymbol{\mu})}\pi^{k/2}[\det\{(2\boldsymbol{V})^{-1}\}]^{-1/2} \tag{A.50}$$

$$= e^{\boldsymbol{t}'\boldsymbol{\mu}+\frac{1}{2}\boldsymbol{t}'\boldsymbol{V}\boldsymbol{t}}. \tag{A.51}$$

Summarizing, if $\boldsymbol{Y} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{V})$, then moment generating function is given by

$$\mathrm{E}\left(e^{\boldsymbol{t}'\boldsymbol{Y}}\right) = e^{\boldsymbol{t}'\boldsymbol{\mu}+\frac{1}{2}\boldsymbol{t}'\boldsymbol{V}\boldsymbol{t}}. \tag{A.52}$$

If we use $i\boldsymbol{t}$ instead of $\boldsymbol{t}$, where i is the complex number such that $i^2 = -1$, we find the characteristic function given by

$$\mathrm{E}\left(e^{i\boldsymbol{t}'\boldsymbol{Y}}\right) = e^{i\boldsymbol{t}'\boldsymbol{\mu}-\frac{1}{2}\boldsymbol{t}'\boldsymbol{V}\boldsymbol{t}}. \tag{A.53}$$

A.5.3 Marginal distributions

Let

$$\boldsymbol{Y} = \begin{pmatrix}\boldsymbol{Y}_1\\ \boldsymbol{Y}_2\end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix}\boldsymbol{\mu}_1\\ \boldsymbol{\mu}2\end{pmatrix}, \begin{pmatrix}\boldsymbol{V}_{11} & \boldsymbol{V}_{12}\\ \boldsymbol{V}_{21} & \boldsymbol{V}_{22}\end{pmatrix}\right), \tag{A.54}$$

where $\boldsymbol{V}_{12} = \boldsymbol{V}_{21}'$, then

$$\boldsymbol{Y}_1 \sim \mathcal{N}(\boldsymbol{\mu}_1,\ \boldsymbol{V}_{11}), \tag{A.55}$$

$$\boldsymbol{Y}_2 \sim \mathcal{N}(\boldsymbol{\mu}_2,\ \boldsymbol{V}_{22}), \tag{A.56}$$

since, for example,

$$\boldsymbol{Y}_1 = (\boldsymbol{I}, \mathbf{O})\boldsymbol{Y} \sim \mathcal{N}\left((\boldsymbol{I}, \mathbf{O})\boldsymbol{\mu},\ (\boldsymbol{I}, \mathbf{O})\boldsymbol{V}\begin{pmatrix}\boldsymbol{I}\\ \mathbf{O}\end{pmatrix}\right). \tag{A.57}$$

If $\boldsymbol{V}_{21} = \mathbf{O}$, then $\boldsymbol{Y}_1$ and $\boldsymbol{Y}_2$ are uncorrelated and consequently independent, due to the multivariate normality. In that case the conditional distributions $f_{\boldsymbol{Y}_1|\boldsymbol{Y}_2}(\boldsymbol{y}_1|\boldsymbol{y}_2)$ and $f_{\boldsymbol{Y}_2|\boldsymbol{Y}_1}(\boldsymbol{y}_2|\boldsymbol{y}_1)$ are equal to the marginal distributions of $\boldsymbol{Y}_1$ and $\boldsymbol{Y}_2$, respectively.

In general, when elements of $\boldsymbol{Y}_1$ are correlated with elements of $\boldsymbol{Y}_2$, i.e. $\mathrm{E}\{(\boldsymbol{Y}_1 - \boldsymbol{\mu}_1)(\boldsymbol{Y}_2 - \boldsymbol{\mu}_2)'\} = \boldsymbol{V}_{12} \neq \mathbf{O}$, then the conditional distributions are different from the marginal distributions. In other words, knowledge of $\boldsymbol{Y}_2$ would provide information about $\boldsymbol{Y}_1$, and conversely, knowledge of $\boldsymbol{Y}_1$ would provide information about $\boldsymbol{Y}_2$.

A.5.4 Conditional distributions

Let

$$\boldsymbol{Y} = \begin{pmatrix}\boldsymbol{Y}_1\\ \boldsymbol{Y}_2\end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix}\boldsymbol{\mu}_1\\ \boldsymbol{\mu}2\end{pmatrix}, \begin{pmatrix}\boldsymbol{V}_{11} & \boldsymbol{V}_{12}\\ \boldsymbol{V}_{21} & \boldsymbol{V}_{22}\end{pmatrix}\right), \tag{A.58}$$

then

$$\boldsymbol{Y}_1|\boldsymbol{Y}_2 = \boldsymbol{y}_2 \sim \mathcal{N}(\boldsymbol{\mu}_{1.2},\ \boldsymbol{V}_{1.2}) \tag{A.59}$$

where

$$\boldsymbol{\mu}_{1.2} = \boldsymbol{\mu}_1 + \boldsymbol{V}_{12}\boldsymbol{V}_{22}^{-1}(\boldsymbol{y}_2 - \boldsymbol{\mu}_2) \tag{A.60}$$

$$\boldsymbol{V}_{1.2} = \boldsymbol{V}_{11} - \boldsymbol{V}_{12}\boldsymbol{V}_{22}^{-1}\boldsymbol{V}_{21} \tag{A.61}$$

$$= \left\{(\boldsymbol{I}, \mathbf{O})\boldsymbol{V}^{-1}\begin{pmatrix}\boldsymbol{I}\\ \mathbf{O}\end{pmatrix}\right\}^{-1}. \tag{A.62}$$

Notice $\boldsymbol{\mu}_{1.2}$ depends linearly on $\boldsymbol{y}_2$, and $\boldsymbol{V}_{1.2}$ does not depend on $\boldsymbol{y}_2$. Similar expressions hold for $\boldsymbol{\mu}_{2.1}$ and $\boldsymbol{V}_{2.1}$.

To prove these results, let's write

$$\begin{pmatrix}\boldsymbol{Y}_1\\ \boldsymbol{Y}_2\end{pmatrix} = \begin{pmatrix}\boldsymbol{\mu}_1\\ \boldsymbol{\mu}_2\end{pmatrix} + \begin{pmatrix}\boldsymbol{A}_1'\\ \boldsymbol{A}_2'\end{pmatrix}\boldsymbol{Z}, \tag{A.63}$$

where $\boldsymbol{Z} \sim \mathcal{N}(\mathbf{0},\ \boldsymbol{I})$ and $\boldsymbol{A}_i'\boldsymbol{A}_j = \boldsymbol{V}_{ij}$, $i, j = 1, 2$.

Decompose $\boldsymbol{A}_1$ into two orthogonal parts

$$\boldsymbol{A}_1 = \boldsymbol{P}_{\boldsymbol{A}_2}\boldsymbol{A}_1 + (\boldsymbol{I} - \boldsymbol{P}_{\boldsymbol{A}_2})\boldsymbol{A}_1, \tag{A.64}$$

where $\boldsymbol{P}_{\boldsymbol{A}_2} = \boldsymbol{A}_2(\boldsymbol{A}_2'\boldsymbol{A}_2)^{-1}\boldsymbol{A}_2'$ is a projection matrix, projecting on the linear subspace spanned by the columns of $\boldsymbol{A}_2$.

Notice that $\boldsymbol{Y}_2$ and $\boldsymbol{A}_1'(\boldsymbol{I} - \boldsymbol{P}_{\boldsymbol{A}_2})\boldsymbol{Z}$ are uncorrelated. That is,

$$\mathrm{E}\left[(\boldsymbol{Y}_2 - \boldsymbol{\mu}_2)\{\boldsymbol{A}_1'(\boldsymbol{I} - \boldsymbol{P}_{\boldsymbol{A}_2})\boldsymbol{Z}\}'\right] = \mathrm{E}\{\boldsymbol{A}_2'\boldsymbol{Z}\boldsymbol{Z}'(\boldsymbol{I} - \boldsymbol{P}_{\boldsymbol{A}_2})\boldsymbol{A}_1\} \tag{A.65}$$

$$= \boldsymbol{A}_2'\,\mathrm{E}(\boldsymbol{Z}\boldsymbol{Z}')(\boldsymbol{I} - \boldsymbol{P}_{\boldsymbol{A}_2})\boldsymbol{A}_1 \tag{A.66}$$

$$= \boldsymbol{A}_2'(\boldsymbol{I} - \boldsymbol{P}_{\boldsymbol{A}_2})\boldsymbol{A}_1 = \mathbf{O}. \tag{A.67}$$

As the elements of $\boldsymbol{Y}_2$ and $\boldsymbol{A}_1'(\boldsymbol{I} - \boldsymbol{P}_{\boldsymbol{A}_2})\boldsymbol{Z}$ have a multivariate normal distribution, they are independent as well. This also holds for functions of the separate parts.

Now $\boldsymbol{Y}_1$ can be decomposed as

$$\boldsymbol{Y}_1 = \boldsymbol{\mu}_1 + \boldsymbol{A}_1'\boldsymbol{Z} \tag{A.68}$$

$$= \boldsymbol{\mu}_1 + \boldsymbol{A}_1\boldsymbol{P}_{\boldsymbol{A}_2}\boldsymbol{Z} + \boldsymbol{A}_1'(\boldsymbol{I} - \boldsymbol{P}_{\boldsymbol{A}_2})\boldsymbol{Z} \tag{A.69}$$

$$= \boldsymbol{\mu}_1 + \boldsymbol{A}_1'\boldsymbol{A}_2(\boldsymbol{A}_2'\boldsymbol{A}_2)^{-1}(\boldsymbol{Y}_2 - \boldsymbol{\mu}_2) + \boldsymbol{A}_1'(\boldsymbol{I} - \boldsymbol{P}_{\boldsymbol{A}_2})\boldsymbol{Z}. \tag{A.70}$$

We find the last two terms on the right hand side are random and independent. The conditional distribution of the third term given $\boldsymbol{Y}_2 = \boldsymbol{y}_2$ is the marginal distribution of the third term, which is given by

$$\boldsymbol{A}_1'(\boldsymbol{I} - \boldsymbol{P}_{\boldsymbol{A}_2})\boldsymbol{Z} \sim \mathcal{N}\left(\mathbf{0},\ \boldsymbol{A}_1'(\boldsymbol{I} - \boldsymbol{P}_{\boldsymbol{A}_2})\boldsymbol{A}_1\right). \tag{A.71}$$

Conditional on $\boldsymbol{Y}_2 = \boldsymbol{y}_2$ the second term on the right hand side is fixed. So, conditional on $\boldsymbol{Y}_2 = \boldsymbol{y}_2$, the covariance matrix of $\boldsymbol{Y}_1$ is given by

$$\boldsymbol{V}_{1.2} = \boldsymbol{A}_1'(\boldsymbol{I} - \boldsymbol{P}_{\boldsymbol{A}_2})\boldsymbol{A}_1 = \boldsymbol{V}_{11} - \boldsymbol{V}_{12}\boldsymbol{V}_{22}^{-1}\boldsymbol{V}_{12}. \tag{A.72}$$

Conditional on $\boldsymbol{Y}_2 = \boldsymbol{y}_2$ the third term has zero expectation. So, given $\boldsymbol{Y}_2 = \boldsymbol{y}_2$, the expectation of $\boldsymbol{Y}_1$ is given by

$$\begin{aligned} \boldsymbol{\mu}_{1.2} &= \boldsymbol{\mu}_1 + \boldsymbol{A}_1'\boldsymbol{A}_2(\boldsymbol{A}_2'\boldsymbol{A}_2)^{-1}(\boldsymbol{y}_2 - \boldsymbol{\mu}_2) && \text{(A.73)} \\ &= \boldsymbol{\mu}_1 + \boldsymbol{V}_{12}\boldsymbol{V}_{22}^{-1}(\boldsymbol{y}_2 - \boldsymbol{\mu}_2). && \text{(A.74)} \end{aligned}$$

A.5.5 Mahalanobis distance

The Euclidean norm is given by $\|\boldsymbol{a}\| = \sqrt{\boldsymbol{a}'\boldsymbol{a}}$.

The Mahalanobis norm is given $\|\boldsymbol{a}\|_{\boldsymbol{V}} = \sqrt{\boldsymbol{a}'\boldsymbol{V}^{-1}\boldsymbol{a}}$, where $\boldsymbol{V} > 0$. The so-called metric is based on the p.d. matrix $\boldsymbol{V}$.

So the Mahalanobis distance between $\boldsymbol{a}_1$ and $\boldsymbol{a}_2$ in the metric $\boldsymbol{V}$ is given by

$$\sqrt{(\boldsymbol{a}_1 - \boldsymbol{a}_2)\boldsymbol{V}^{-1}(\boldsymbol{a}_1 - \boldsymbol{a}_2)}. \tag{A.75}$$

The ellipsoid $\{\boldsymbol{y} \mid (\boldsymbol{y} - \boldsymbol{\mu})\boldsymbol{V}^{-1}(\boldsymbol{y} - \boldsymbol{\mu}) = c^2\}$ has points $\boldsymbol{y}$ with the same mahalanobis distance $\|\boldsymbol{y} - \boldsymbol{\mu}\|_{\boldsymbol{V}} = c$ to the point $\boldsymbol{\mu}$.

A.5.6 The Chi-squared distribution

The family of Chi-squared distributions is indexed by the degrees of freedom.

Let $\boldsymbol{Z} \sim \mathcal{N}(\boldsymbol{0},\ \boldsymbol{I}_k)$, so $\boldsymbol{Z}$ is a random $k \times 1$ vector, then

$$U_k = \boldsymbol{Z}'\boldsymbol{Z} = \sum_{i=1}^{k} Z_i^2 \sim \chi_k^2. \tag{A.76}$$

It is said to have k degrees of freedom.[1] We find

$$\begin{aligned} \mathrm{E}(U_k) &= k, && \text{(A.77)} \\ \mathrm{var}(U_k) &= 2k, && \text{(A.78)} \end{aligned}$$

since

$$\begin{aligned} \mathrm{var}(U_k) &= k\,\mathrm{var}(U_1) && \text{(A.79)} \\ &= k\left\{\mathrm{E}(Z_1^4) - \mathrm{E}(Z_1^2)^2\right\} && \text{(A.80)} \\ &= k(3-1) = 2k. && \text{(A.81)} \end{aligned}$$

[1] A Chi-squared distribution is a Gamma distribution $U_k \sim \mathrm{Gamma}(k/2, 1/2)$.

Let

$$\boldsymbol{Z} = \begin{pmatrix} \boldsymbol{Z}_1 \\ \boldsymbol{Z}_2 \end{pmatrix},$$

where $\boldsymbol{Z}_1$ and $\boldsymbol{Z}_2$ are k_1 and k_2 dimensional, then $U_{k_i} = \boldsymbol{Z}_i'\boldsymbol{Z}_i \sim \chi^2_{k_i}$, $i = 1, 2$ and the two terms are independent, so

$$U_k = U_{k_1} + U_{k_2} \sim \chi^2_k,$$

where $k = k_1 + k_2$.

We discuss some material from **Section A.5.8** *as well.*

Let $\boldsymbol{P}$ be a $k \times k$ orthonormal matrix, so $\boldsymbol{P}'\boldsymbol{P} = \boldsymbol{P}\boldsymbol{P}' = \boldsymbol{I}_k$, and let it be partitioned $\boldsymbol{P} = (\boldsymbol{P}_1, \boldsymbol{P}_2)$, where $\boldsymbol{P}_1$ is $k \times k_1$ and $\boldsymbol{P}$ is $k \times k_2$.

Notice that $\boldsymbol{P}_1\boldsymbol{P}_1'$ and $\boldsymbol{P}_2\boldsymbol{P}_2'$ are projection matrices. They have ranks equal to k_1 and k_2, respectively.

Let $\boldsymbol{X} = \boldsymbol{P}\boldsymbol{Z}$. As

$$\boldsymbol{X} = \begin{pmatrix} \boldsymbol{X}_1 \\ \boldsymbol{X}_2 \end{pmatrix} = \boldsymbol{P}'\boldsymbol{Z} = \begin{pmatrix} \boldsymbol{P}_1'\boldsymbol{Z} \\ \boldsymbol{P}_2'\boldsymbol{Z} \end{pmatrix} \sim \mathcal{N}(\boldsymbol{0},\ \boldsymbol{I}_k), \tag{A.82}$$

we have

$$\boldsymbol{X}'\boldsymbol{X} = \boldsymbol{X}_1'\boldsymbol{X}_1 + \boldsymbol{X}_2'\boldsymbol{X}_2, \tag{A.83}$$

and

(i) $\boldsymbol{Z}'\boldsymbol{P}_1\boldsymbol{P}_1'\boldsymbol{Z} = \boldsymbol{X}_1'\boldsymbol{X}_1 \sim \chi^2_{k_1}$,

(ii) $\boldsymbol{Z}'\boldsymbol{P}_2\boldsymbol{P}_2'\boldsymbol{Z} = \boldsymbol{X}_2'\boldsymbol{X}_2 \sim \chi^2_{k_2}$,

(iii) $\boldsymbol{Z}'\boldsymbol{P}_1\boldsymbol{P}_1'\boldsymbol{Z}$ and $\boldsymbol{Z}'\boldsymbol{P}_2\boldsymbol{P}_2'\boldsymbol{Z}$ are independent (since $\boldsymbol{P}_1'\boldsymbol{P}_2 = \mathbf{O}$).

Such results can be formulated in a theorem (Fisher-Cochrane) that does not use projection matrices explicitly. However we prove it using the following Lemma.

Lemma 1 *Let $\boldsymbol{A}_i$ be $k \times k$, $\boldsymbol{A}_i \geq 0$, have ranks* $\mathrm{rank}(\boldsymbol{A}_i) = r_i$, *$i = 1, \ldots, m$, such that*

$$\sum_{i=1}^{m} = \boldsymbol{A}_i = \boldsymbol{I}_k,$$

then the following implication holds. If $\sum_{i=1}^m r_i = k$, then $\boldsymbol{A}_i$, $i = 1, \ldots, m$ are projection matrices and $\boldsymbol{A}_i\boldsymbol{A}_j = \mathbf{O}$ for $i \neq j$.

Proof

As $\boldsymbol{A}_i \geq 0$, we may write $\boldsymbol{A}_i = \boldsymbol{B}_i\boldsymbol{B}_i'$ for matrices $\boldsymbol{B}_i$ of order $k \times r_i$ that can be found using the SVD of $\boldsymbol{A}_i$, $i = 1, \ldots, m$. Let $\boldsymbol{B} = (\boldsymbol{B}_1, \ldots, \boldsymbol{B}_m)$, which is $k \times \sum_{i=1}^m r_i$. Consequently,

$$\boldsymbol{B}\boldsymbol{B}' = \sum_{i=1}^{m} \boldsymbol{B}_i\boldsymbol{B}_i' = \sum_{i=1}^{m} \boldsymbol{A}_i = \boldsymbol{I}_k. \tag{A.84}$$

So, the rows of $\boldsymbol{B}$ are orthonormal. The condition $\sum_{i=1}^m r_i = k$, says $\boldsymbol{B}$ is square. In that case the columns of $\boldsymbol{B}$ are orthonormal as well. So, the matrices $\boldsymbol{B}_i\boldsymbol{B}_i'$, $i = 1, \ldots, m$ are projection matrices. q.e.d.

Theorem 1 *(Fisher-Cochrane)*
Let $\boldsymbol{Z}$ be a random k vector with a multivariate standard normal distribution, $\boldsymbol{Z} \sim \mathcal{N}(\boldsymbol{0},\ \boldsymbol{I}_k)$, and let $\boldsymbol{A}_i$ be $k \times k$, $\boldsymbol{A}_i \geq 0$, with ranks $\text{rank}(\boldsymbol{A}_i) = r_i$, *$i = 1, \ldots, m$, such that*

$$\sum_{i=1}^m = \boldsymbol{A}_i = \boldsymbol{I}_k,$$

then the following two statements are equivalent:

(i) $\boldsymbol{Z}'\boldsymbol{A}_i\boldsymbol{Z}$, *$i = 1, \ldots, m$, are independent and* $\boldsymbol{Z}'\boldsymbol{A}_i\boldsymbol{Z} \sim \chi^2_{r_i}$,

(ii) $\sum_{i=1}^m r_i = k$.

Proof
(ii) $\Rightarrow$ (i): If (ii) holds then by Lemma 1 the matrices $\boldsymbol{A}_i$ are projection matrices with $\boldsymbol{A}_i\boldsymbol{A}_j = \boldsymbol{O}$, so the results follows as discussed above.
(i) $\Rightarrow$ (ii): If $\boldsymbol{Z}'\boldsymbol{A}_i\boldsymbol{Z}$, $i = 1, \ldots, m$, are independent and $\boldsymbol{Z}'\boldsymbol{A}_i\boldsymbol{Z} \sim \chi^2_{r_i}$, then

$$\sum_{i=1}^m \boldsymbol{Z}'\boldsymbol{A}_i\boldsymbol{Z} \sim \chi^2_{\sum_i r_i}. \tag{A.85}$$

As

$$\sum_{i=1}^m \boldsymbol{Z}\boldsymbol{A}_i\boldsymbol{Z} = \boldsymbol{Z}'\left(\sum_{i=1}^m \boldsymbol{A}_i\right)\boldsymbol{Z} = \boldsymbol{Z}'\boldsymbol{Z} \sim \chi^2_k, \tag{A.86}$$

we find $\sum_{i=1}^m r_i = k$, which is (ii). q.e.d.

Frequently, we will apply the following reasoning in case $\boldsymbol{Z}$ has a multivariate standard normal distribution. If $\boldsymbol{P}_A$ and $\boldsymbol{P}_B$ are projection matrices with ranks r_A and r_B, respectively, and $\boldsymbol{P}_A\boldsymbol{P}_B = \boldsymbol{O}$, then

(i) $\boldsymbol{Z}'\boldsymbol{P}_A\boldsymbol{Z} \sim \chi^2_A$,

(ii) $\boldsymbol{Z}'\boldsymbol{P}_B\boldsymbol{Z} \sim \chi^2_B$,

(iii) they are independent.

We also find $\boldsymbol{P}_A + \boldsymbol{P}_B$ is a projection matrix with rank $r_A + r_B$ and

$$\boldsymbol{Z}'(\boldsymbol{P}_A + \boldsymbol{P}_B)\boldsymbol{Z} \sim \chi^2_{r_A + r_B}. \tag{A.87}$$

The book discusses the Fisher-Cochrane Theorem for noncentral chi-squared distributions, which we discuss later.

Remark 5
Notation
We will write $X \sim \sigma^2\chi^2_{df}$ *if* $X/\sigma^2 \sim \chi^2_{df}$.

Remark 6
Hint for Exercise 4
If $\boldsymbol{Y} \sim \mathcal{N}(\boldsymbol{\mu},\ \boldsymbol{V})$, *then* $\boldsymbol{Z} = \boldsymbol{V}^{-1/2}(\boldsymbol{Y} - \boldsymbol{\mu}) \sim \mathcal{N}(\boldsymbol{0},\ \boldsymbol{I})$. *If the* $k \times r$ *matrix* $\boldsymbol{B}$ *has f.c.r. then it can be shown that*

$$(\boldsymbol{Y} - \boldsymbol{\mu})'\boldsymbol{B}(\boldsymbol{B}'\boldsymbol{V}\boldsymbol{B})^{-1})\boldsymbol{B}'(\boldsymbol{Y} - \boldsymbol{\mu}) \sim \chi^2_r. \tag{A.88}$$

Exercise 4 asks to show it.

Remark 7
Hint for Exercise 5
Let $\boldsymbol{\Omega} > 0$ *and let the* $n \times k$ *matrix* $\boldsymbol{X}$ *have f.c.r., then Exercise 5 asks to show that*

$$\boldsymbol{X}'\boldsymbol{\Omega}^{-1}\boldsymbol{X} \leq (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\Omega}\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}. \tag{A.89}$$

You might want to use the property that if $\boldsymbol{P}$ *is a projection matrix, then* $0 \leq \boldsymbol{P} \leq \boldsymbol{I}$. *So, for example,* $\boldsymbol{A}'\boldsymbol{P}\boldsymbol{A} \leq \boldsymbol{A}'\boldsymbol{A}$.

Remark 8
Asymptotic result
Notice that if $U_k \sim \chi^2_k$, *then*

$$\mathrm{E}(U_k/k) = 1, \tag{A.90}$$

$$\mathrm{var}(U_k/k) = 2/k \to 0, \qquad \textit{as} \ \ k \to \infty. \tag{A.91}$$

So, U_k/k *converges in this sense to 1 as* $k \to \infty$. *It is called convergence in quadratic mean,* $\lim_{k\to\infty} \mathrm{E}\left\{(U_k/k - 1)^2\right\} = 0$.

A.5.7 The t and F distributions

The t and F distributions form families of distributions, just as the family of Chi-squared distributions, which was indexed by the degrees of freedom.

Let the scalar random variable Z have a standard normal distribution, $Z \sim \mathcal{N}(0,\ 1)$ and, independently, $U_k \sim \chi^2_k$, then we say

$$T = \frac{Z}{\sqrt{U_k/k}} \sim t_k, \tag{A.92}$$

i.e. it has a t distribution with k degrees of freedom.

It has some properties:

1. the distribution is symmetric about zero, i.e. $f_T(t) = f_T(-t)$, where f_T denotes the density of T;

2. the expectation of the ith moment $\mathrm{E}(T^i)$ exists for $i \leq k-1$;
3. for $k = 1$ the distribution is equal to the Cauchy distribution;
4. as $k \to \infty$, the t_k distribution converges to the standard normal distribution.

Let $U_k \sim \chi^2_k$ and, independently, $V_m \sim \chi^2_m$, then we say

$$F^* = \frac{U_k/k}{V_m/m} \sim F_{k,m}, \tag{A.93}$$

i.e. it has an F distribution with k and m degrees of freedom.

Some properties are given by:

1. if $T \sim t_k$, then $T^2 \sim F_{1,k}$;
2. if $T^* \sim F_{k,m}$, then $T^{*-1} \sim F_{m,k}$;
3. as $m \to \infty$, the distribution of kF^* converges to the χ^2_k distribution.

A.5.8 Distribution of the sample mean and variance

Let $Y_1, \ldots, Y_n$ be independent and identically distributed (i.i.d.) random variables with a normal distribution $\mathcal{N}(\mu,\ \sigma^2)$.

For example we measure the IQ of a random sample of n persons. We assume it is normally distributed, or we approximate it by a normal distribution.

The sample mean and sample variance are given by

$$\overline{Y}_n = \frac{1}{n}\sum_{i=1}^{n} Y_i, \tag{A.94}$$

$$S_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \overline{Y})^2. \tag{A.95}$$

In case of normality, as we assume it, they are independent and

$$\overline{Y}_n \sim \mathcal{N}(\mu,\ \sigma^2/n), \tag{A.96}$$

$$S_n^2 \sim \left(\frac{\sigma^2}{n-1}\right)\chi^2_{n-1}. \tag{A.97}$$

To verify this, let $\boldsymbol{\iota}_n$ be a vector of n ones, and let

$$\boldsymbol{P}_{\boldsymbol{\iota}_n} = \boldsymbol{\iota}_n(\boldsymbol{\iota}_n'\boldsymbol{\iota}_n)^{-1}\boldsymbol{\iota}_n' = \frac{\boldsymbol{\iota}_n\boldsymbol{\iota}_n'}{\boldsymbol{\iota}_n'\boldsymbol{\iota}_n} = \frac{\boldsymbol{\iota}_n\boldsymbol{\iota}_n'}{n},$$

Let $\boldsymbol{Y} = (Y_1, \ldots, Y_n)'$, then

$$\overline{Y}_n = (\boldsymbol{\iota}_n'\boldsymbol{\iota}_n)^{-1}\boldsymbol{\iota}_n'\boldsymbol{Y} = \frac{\boldsymbol{\iota}_n'\boldsymbol{Y}}{n}, \tag{A.98}$$

$$S_n^2 = \frac{\boldsymbol{Y}'(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{\iota}_n})\boldsymbol{Y}}{\mathrm{tr}(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{\iota}_n})} = \frac{\boldsymbol{Y}'(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{\iota}_n})\boldsymbol{Y}}{n-1}. \tag{A.99}$$

As,

$$\boldsymbol{Y} \sim \mathcal{N}(\boldsymbol{\iota}_n \mu, \sigma^2 \boldsymbol{I}_n), \tag{A.100}$$

we find

$$\overline{Y}_n \sim \mathcal{N}(\mu,\ \sigma^2/n), \tag{A.101}$$

since

$$\mathrm{E}\left(\frac{\boldsymbol{\iota}_n' \boldsymbol{Y}}{n}\right) = \mu \frac{\boldsymbol{\iota}_n' \boldsymbol{\iota}_n}{n} = \mu, \tag{A.102}$$

$$\mathrm{var}\left(\frac{\boldsymbol{\iota}_n \boldsymbol{Y}}{n}\right) = \frac{1}{n^2}\,\mathrm{var}(\boldsymbol{\iota}_n \boldsymbol{Y}) = \frac{1}{n^2}\sigma^2 \boldsymbol{\iota}_n' \boldsymbol{I}_n \boldsymbol{\iota}_n = \frac{\sigma^2}{n}. \tag{A.103}$$

Remark 9
Remember
If ε_i, $i = 1, \ldots, n$ are i.i.d. and $\mathrm{var}(\varepsilon_i) = \sigma^2$, then

$$\mathrm{var}\left(\sum_{i=1}^{n} \varepsilon_i\right) = n\sigma^2, \qquad \mathrm{var}\left(\frac{1}{n}\sum_{i=1}^{n} \varepsilon_i\right) = \frac{n}{n^2}\sigma^2 = \frac{\sigma^2}{n}. \tag{A.104}$$

Furthermore,

$$S_n^2 = \frac{\boldsymbol{Y}'(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{\iota}_n})\boldsymbol{Y}}{n-1} = \frac{(\boldsymbol{Y} - \boldsymbol{\iota}_n \mu)'(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{\iota}_n})(\boldsymbol{Y} - \boldsymbol{\iota}_n \mu)}{n-1}, \tag{A.105}$$

since $(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{\iota}_n})\boldsymbol{\iota}_n = \boldsymbol{0}$.

So,

$$\frac{(n-1)}{\sigma^2} S_n^2 = \left(\frac{\boldsymbol{Y} - \boldsymbol{\iota}_n \mu}{\sigma}\right)' (\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{\iota}_n}) \left(\frac{\boldsymbol{Y} - \boldsymbol{\iota}_n \mu}{\sigma}\right). \tag{A.106}$$

As $\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{\iota}_n}$ is a projection matrix with rank $k-1$, and since

$$\frac{\boldsymbol{Y} - \boldsymbol{\iota}_n \mu}{\sigma} \sim \mathcal{N}(\boldsymbol{0},\ \boldsymbol{I}_n), \tag{A.107}$$

we find

$$\frac{(n-1)}{\sigma^2} S_n^2 \sim \chi^2_{n-1}. \tag{A.108}$$

Moreover, S_n^2 and $\overline{\boldsymbol{Y}}_n$ are independent since $\overline{\boldsymbol{Y}}_n$ is a function of $\boldsymbol{\iota}_n' \boldsymbol{Y}$ and S_n^2 is a function of $(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{\iota}_n})\boldsymbol{Y}$. These terms are uncorrelated,

$$\begin{aligned}
\mathrm{Cov}\left\{(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{\iota}_n})\boldsymbol{Y}, \boldsymbol{\iota}_n' \boldsymbol{Y}\right\} &= (\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{\iota}_n})\,\mathrm{Cov}(\boldsymbol{Y}, \boldsymbol{Y}')\boldsymbol{\iota}_n && \text{(A.109)}\\
&= (\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{\iota}_n})\,\mathrm{E}\left\{(\boldsymbol{Y} - \boldsymbol{\iota}_n \mu)(\boldsymbol{Y} - \boldsymbol{\iota}_n \mu)'\right\}\boldsymbol{\iota}_n && \text{(A.110)}\\
&= (\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{\iota}_n})\,\mathrm{Var}(\boldsymbol{Y})\boldsymbol{\iota}_n && \text{(A.111)}\\
&= \sigma^2(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{\iota}_n})\boldsymbol{I}_n \boldsymbol{\iota}_n = \sigma^2(\boldsymbol{I}_n - \boldsymbol{P}_{\boldsymbol{\iota}_n})\boldsymbol{\iota}_n = \boldsymbol{0}, && \text{(A.112)}
\end{aligned}$$

and due to the multivariate normality, the terms are independent as well. So we have proven the three statements at the beginning of this section.

As

(i) $\overline{Y}_n \sim \mathcal{N}(\mu,\ \sigma^2/n)$,

(ii) $\frac{(n-1)}{\sigma^2} S_n^2 \sim \chi^2_{n-1}$,

(iii) $\overline{Y}_n$ and $\frac{(n-1)}{\sigma^2} S_n^2$ are independent,

we could construct a ratio of a numerator given by

$$\sqrt{\frac{n}{\sigma^2}}(\overline{Y}_n - \mu) \sim \mathcal{N}(0,\ 1), \tag{A.113}$$

and a denominator given by the square root of

$$\frac{(n-1)}{\sigma^2} S_n^2/(n-1) \sim \frac{\chi^2_{n-1}}{n-1}, \tag{A.114}$$

to form a random variable with a t_{n-1} distribution

$$T = \frac{\sqrt{\frac{n}{\sigma^2}}(\overline{Y}_n - \mu)}{\sqrt{\frac{(n-1)}{\sigma^2} S_n^2/(n-1)}} = \sqrt{n}\left(\frac{\overline{Y}_n - \mu}{S_n}\right) \sim t_{n-1}. \tag{A.115}$$

If μ were known, T could be computed based on the data.

If μ is unknown, we might hypothesize that $\mu = \mu^*$ holds for some particular value of μ^*. If that hypothesis is true, then

$$T = \sqrt{n}\left(\frac{\overline{Y}_n - \mu^*}{S_n}\right) \tag{A.116}$$

should be considered a random drawing from the t_{n-1} distribution. We could test that.

A.5.9 The noncentral Chi-squared distribution

Let $\boldsymbol{Y} \sim \mathcal{N}(\boldsymbol{\mu},\ \boldsymbol{I}_k)$, then we say

$$U_k = \boldsymbol{Y}'\boldsymbol{Y} \sim \chi^2_k(\boldsymbol{\mu}'\boldsymbol{\mu}). \tag{A.117}$$

It has a noncentral Chi-squared distribution with k degrees of freedom and noncentrality parameter $\delta = \boldsymbol{\mu}'\boldsymbol{\mu}$.

Notice it is easy to make tables of the distributions, $\mathrm{P}(T \leq x)$ for the χ^2, t and F distributions, since they only depend on the number(s) of degrees of freedom.

For the noncentral distributions we would also need to know the noncentrality parameter.[2]

By saying $U_k \sim \chi_k^2(\boldsymbol{\mu}'\boldsymbol{\mu})$ we made the claim that the distribution depends only on $\boldsymbol{\mu}'\boldsymbol{\mu}$. We need not the whole vector $\boldsymbol{\mu}$, but only its length. This claim can be proved. We have

$$U_k = \boldsymbol{Y}'\boldsymbol{Y} = \boldsymbol{Y}'(\boldsymbol{I}_k - \boldsymbol{P}_{\boldsymbol{\mu}})\boldsymbol{Y} + \boldsymbol{Y}'\boldsymbol{P}_{\boldsymbol{\mu}}\boldsymbol{Y}, \tag{A.118}$$

where $\boldsymbol{P}_{\boldsymbol{\mu}} = \boldsymbol{\mu}(\boldsymbol{\mu}'\boldsymbol{\mu})^{-1}\boldsymbol{\mu}'$, which is defined for all $\boldsymbol{\mu} \neq \boldsymbol{0}$. The first term on the right hand side does not depend on $\boldsymbol{\mu}$ as

$$\boldsymbol{Y}'(\boldsymbol{I}_k - \boldsymbol{P}_{\boldsymbol{\mu}})\boldsymbol{Y} \sim \chi_{k-1}^2. \tag{A.119}$$

The second term $\boldsymbol{Y}'\boldsymbol{P}_{\boldsymbol{\mu}}\boldsymbol{Y}$ is independent of the first term and it equals

$$\boldsymbol{Y}'\boldsymbol{P}_{\boldsymbol{\mu}}\boldsymbol{Y} = \left(\frac{\boldsymbol{\mu}'\boldsymbol{Y}}{\sqrt{\boldsymbol{\mu}'\boldsymbol{\mu}}}\right)^2. \tag{A.120}$$

As

$$\frac{\boldsymbol{\mu}'\boldsymbol{Y}}{\sqrt{\boldsymbol{\mu}'\boldsymbol{\mu}}} \sim \mathcal{N}(\sqrt{\boldsymbol{\mu}'\boldsymbol{\mu}},\ 1), \tag{A.121}$$

we find $\boldsymbol{Y}'\boldsymbol{P}_{\boldsymbol{\mu}}\boldsymbol{Y}$ and, consequently, $\boldsymbol{Y}'\boldsymbol{Y}$ depend only on $\boldsymbol{\mu}$ through $\sqrt{\boldsymbol{\mu}'\boldsymbol{\mu}}$.

So, although we started with a vector $\boldsymbol{\mu}$, we find the distribution of $\boldsymbol{Y}'\boldsymbol{Y}$ is only affected by $\delta = \boldsymbol{\mu}'\boldsymbol{\mu}$.

Theorem A.5.4 If the random k vector $\boldsymbol{Y}$ has a normal distribution, $\boldsymbol{Y} \sim \mathcal{N}(\boldsymbol{\mu},\ \boldsymbol{V})$, then

$$\boldsymbol{Y}'\boldsymbol{V}^{-1}\boldsymbol{Y} \sim \chi_k^2(\boldsymbol{\mu}\boldsymbol{V}^{-1}\boldsymbol{\mu}). \tag{A.122}$$

The proof follows immediately from

$$\boldsymbol{V}^{-1/2}\boldsymbol{Y} \sim \mathcal{N}(\boldsymbol{V}^{-1/2}\boldsymbol{\mu},\ \boldsymbol{I}_k). \tag{A.123}$$

The book also formulates the Fisher-Cochrane Theorem in terms of the noncentral Chi-squared distribution.

[2]We say F quantities have a noncentral F distribution if the Chi-squared distributed random variable in the numerator has a noncentrality parameter different from zero.